

1. Theory of Interest

Accumulation Functions

Simple:  $a(t) = 1 + rt$     Compound:  $a(t) = (1 + r)^t$  [default]

Continuous:  $a(t) = e^{\delta t}$

Accumulated value of principal  $P$ :  $A(t) = P \cdot a(t)$

Frequency of Compounding

Nominal rate  $r^{(p)}$  paid  $p$  times/yr:

$$a(t) = \left(1 + \frac{r^{(p)}}{p}\right)^{pt}$$

Effective annual rate:

$$r_e = \left(1 + \frac{r^{(p)}}{p}\right)^p - 1$$

Equivalence of two nominal rates:

$$\left(1 + \frac{r^{(p)}}{p}\right)^p = \left(1 + \frac{r^{(q)}}{q}\right)^q$$

$r_e > r^{(p)}$  for any  $p > 1$ . Continuous compounding maximises  $r_e$  for a given nominal rate.

Force of Interest

$$\delta(t) = \frac{a'(t)}{a(t)} = \frac{d}{dt} \ln a(t)$$

Constant  $\delta$ :  $a(t) = e^{\delta t}$ ,  $\delta = \ln(1 + r_e)$ ,  $r_e = e^{\delta} - 1$ .

Variable force of interest:

$$a(t) = \exp\left(\int_0^t \delta(u) du\right) \quad a(s, t) = \exp\left(\int_s^t \delta(u) du\right)$$

Key Conversions (quick ref)

| Convert                       | Formula                                        |
|-------------------------------|------------------------------------------------|
| $r^{(p)} \rightarrow r_e$     | $(1 + r^{(p)}/p)^p - 1$                        |
| $r_e \rightarrow r^{(p)}$     | $p[(1 + r_e)^{1/p} - 1]$                       |
| $r_e \rightarrow \delta$      | $\ln(1 + r_e)$                                 |
| $\delta \rightarrow r_e$      | $e^{\delta} - 1$                               |
| $r^{(p)} \rightarrow r^{(q)}$ | equate $(1 + r^{(p)}/p)^p = (1 + r^{(q)}/q)^q$ |

2. Present Value & Cash Flows

Present Value (single payment)

$$PV = \frac{c}{a(t)} = \frac{c}{(1 + r)^t}$$

Discount factor:  $v = \frac{1}{1+r}$ , so  $PV = c \cdot v^t$

PV of Cash Flow Stream

$C = \{(c_i, t_i)\}_{i=1}^n$ :

$$PV(C) = \sum_{i=1}^n \frac{c_i}{(1 + r)^{t_i}}$$

Time value at  $t$ :  $TV_t(C) = PV(C) \cdot (1 + r)^t$

Annuities

Let  $r$  = effective rate per period,  $A$  = payment per period,  $n$  = number of periods.

Ordinary annuity (payment at end of each period):

$$PV = \frac{A}{r} \left[1 - \frac{1}{(1 + r)^n}\right]$$

Annuity-due (payment at start of each period):

$$PV = \frac{A(1 + r)}{r} \left[1 - \frac{1}{(1 + r)^n}\right]$$

$\Rightarrow PV(\text{due}) = PV(\text{ordinary}) \times (1 + r)$

Perpetuity (ordinary, payments at end):

$$PV = \frac{A}{r}$$

Perpetuity-due (payments at start):

$$PV = \frac{A(1 + r)}{r}$$

Future value (ordinary annuity at  $t = n$ ):

$$FV = \frac{A}{r} [(1 + r)^n - 1]$$

Deferred annuity ( $n$  payments, first at  $t = m + 1$ ):

$$PV = \frac{A}{r} \left[1 - \frac{1}{(1 + r)^n}\right] \cdot \frac{1}{(1 + r)^m}$$

Discount ordinary annuity PV back by  $m$  periods.

Rate mismatch: Always convert nominal rate to effective rate per payment period before applying annuity formula. E.g. 6% p.a. convertible semi-annually  $\rightarrow$  effective monthly rate  $= (1 + 0.03)^{1/6} - 1$ .

NPV & IRR Decision Rules

NPV criterion: Invest if  $NPV > 0$ . Prefer A over B if  $NPV(A) > NPV(B)$ .

IRR criterion: Invest if  $IRR >$  required return. Prefer A over B if  $IRR(A) > IRR(B)$ .

IRR is  $r$  s.t.  $PV(C) = 0$ . Uniqueness: if  $c_0 < 0$ ,  $c_k \geq 0 \forall k \geq 1$ ,  $\sum c_k > 0 \Rightarrow$  unique IRR.

Newton–Raphson:  $r_{n+1} = r_n - f(r_n)/f'(r_n)$

Growing Annuity & Perpetuity

First payment  $A$  at  $t = 1$ , grows at rate  $g$ /period,  $n$  periods:

$$PV = \frac{A}{r - g} \left[1 - \left(\frac{1 + g}{1 + r}\right)^n\right] \quad (r \neq g) \quad r = g : \frac{nA}{1 + r}$$

Growing perpetuity ( $r > g$ ):  $PV = A/(r - g)$ .

Loans, Refinancing & Last Payment

Loan  $L$ ,  $n$  payments  $A$ , rate  $r$ /period:  $L = \frac{A}{r} [1 - (1 + r)^{-n}]$ ,

$$A = \frac{Lr}{1 - (1 + r)^{-n}}$$

Balance after  $m$  payments:  $L_m^{\text{bal}} = \frac{A}{r} [1 - (1 + r)^{-(n-m)}]$  (prospective)

$L_m^{\text{bal}} = L(1 + r)^m - \frac{A}{r} [(1 + r)^m - 1]$  (retrospective)

Irregular last payment ( $n$  non-integer): make  $[n]$  full payments of  $A$  plus final smaller payment  $B$ .

•  $B$  at  $t = [n]$ : set  $L = PV$  of  $[n] - 1$  payments of  $A$  plus  $B$  at  $t = [n]$ .

•  $B$  at  $t = [n] + 1$ : use  $L_{[n]}^{\text{bal}}$ , then  $B = L_{[n]}^{\text{bal}}(1 + r)$ .

3. Bonds

Notation

|           |                                  |
|-----------|----------------------------------|
| $F$       | face value (= $R$ unless stated) |
| $R$       | redemption value                 |
| $c$       | nominal coupon rate (annual)     |
| $m$       | coupon payments per year         |
| $n$       | total coupon payments            |
| $\lambda$ | nominal bond yield (annual)      |
| $P$       | bond price                       |

Bond Pricing Formula

$$P = \frac{R}{(1 + \lambda/m)^n} + \sum_{i=1}^n \frac{cF/m}{(1 + \lambda/m)^i}$$

$$= F \left[ \frac{1}{(1 + \lambda/m)^n} + \frac{c/m}{\lambda/m} \left(1 - \frac{1}{(1 + \lambda/m)^n}\right) \right] \quad (F = R)$$

Premium/discount form ( $F = R$ ):  $P = F + F \cdot \frac{c - \lambda}{\lambda} [1 - (1 + \lambda/m)^{-n}]$

$c > \lambda \Rightarrow P > F$  (premium);  $c = \lambda \Rightarrow P = F$  (par);  $c < \lambda \Rightarrow P < F$  (discount).

Makeham formula (alternative; let  $K = R/(1 + \lambda/m)^n$ ):

$$P = K + \frac{c}{\lambda} (F - K) \quad (F = R \text{ case})$$

Recursive price:  $P_{k+1} = P_k(1 + \lambda/m) - cF/m$  (price after  $k$ -th coupon).

Variable Coupon Bonds

Split into groups with constant coupon rate; price each group as a separate annuity + shared redemption PV.

Exam pattern (Q3 midterm): Bond with different coupon rates for each 5-year block and  $R \neq F$ . Price each block as an annuity discounted at yield  $\lambda$ , then add PV of redemption at maturity.

Determining Bond Yield (Newton–Raphson)

$f(\lambda) = P(\lambda) - P_{\text{market}}$ , iterate  $\lambda_{n+1} = \lambda_n - f(\lambda_n)/f'(\lambda_n)$ .

$$P(\lambda) = \frac{R}{(1 + \lambda/m)^n} + \frac{cF/m}{\lambda/m} \left[1 - \frac{1}{(1 + \lambda/m)^n}\right]$$

$f'(\lambda)$  for annual coupon bond ( $m = 1$ ,  $F = R$ ):

$$f'(\lambda) = - \sum_{i=1}^n \frac{i \cdot cF}{(1 + \lambda)^{i+1}} - \frac{nF}{(1 + \lambda)^{n+1}} = - \frac{D \cdot P(\lambda)}{1 + \lambda}$$

where  $D$  = Macaulay duration. (In general:  $f'(\lambda) = - \frac{D \cdot P}{m(1 + \lambda/m)} \cdot$ )

Start:  $\lambda_0 = c$ . Price–yield: **inverse** & **convex** ( $P \downarrow$  as  $\lambda \uparrow$ ).

Macaulay Duration

$$D = \frac{\sum_{i=1}^n t_i \cdot PV(c_i, t_i)}{PV(C)} = \sum_{i=1}^n w_i t_i, \quad w_i = \frac{PV(c_i)}{PV(C)}$$

Special cases ( $\mu = \lambda/m$ ,  $\gamma = c/m$ ): Zero-coupon:  $D = T$ . Perpetuity ( $m = 1$ ):  $D = (1 + \lambda)/\lambda$ .

**Coupon bond,  $F = R$  (general):** 
$$D = \frac{1 + \mu}{m\mu} - \frac{1 + \mu + n(\gamma - \mu)}{m\gamma[(1 + \mu)^n - 1] + m\mu} \quad (2)$$
  
**At par  $P = F = R$  ( $\mu = \gamma$ ):** 
$$D = \frac{1 + \mu}{m\mu} \left[1 - \frac{1}{(1 + \mu)^n}\right] \quad (3)$$

$t_1 \leq D \leq t_n$ ;  $D$  independent of  $F$  and of yield compounding frequency.

4. Term Structure

Spot Rates & Forward Rates

Spot rate  $s_t$ : effective annual rate from 0 to  $t$ . Accumulation:  $(1 + s_t)^t$ .

Forward rate  $f_{j,k}$ : rate from  $t = j$  to  $t = k$  (annual, same convention as spot).

No-arbitrage relation:

$$(1 + s_k)^k = (1 + s_j)^j (1 + f_{j,k})^{k-j}$$

Note:  $f_{0,t} = s_t$ .

Continuous compounding version:

$$e^{s_k k} = e^{s_j j} \cdot e^{f_{j,k}(k-j)} \Rightarrow f_{j,k} = \frac{s_k k - s_j j}{k - j}$$

Bootstrapping Spot Rates

Use observed coupon bond prices to extract  $s_1, s_2, \dots$  sequentially:

For a coupon bond with annual payments:

$$P = \frac{c_1}{1 + s_1} + \frac{c_2}{(1 + s_2)^2} + \dots + \frac{c_n + F}{(1 + s_n)^n}$$

Solve for the unknown  $s_n$  given  $s_1, \dots, s_{n-1}$  already found.

Exam pattern: Extract  $s_1, s_2$  from zero-coupon bond prices, then use coupon bond price to find  $s_3$ , then compute  $f_{1,3}$ .

Yield Curve Shapes

Normal (upward): economic expansion, low inflation.

Inverted (downward): pre-recession, high inflation.

Flat term structure: all  $s_t$  equal  $\Rightarrow$  force of interest is constant.

5. Asset Returns & Portfolios

Returns

Total return:  $R = P_1/P_0$     Rate of return:  $r = (P_1 - P_0)/P_0$      $R = 1 + r$

Short sell:  $R_{\text{short}} = P_1/P_0$  (same formula – negatives cancel).

Random Returns & Key Identities

Mean:  $\mu_i = E(r_i)$     Variance:  $\sigma_i^2 = \text{Var}(r_i)$

Covariance:  $\sigma_{ij} = \text{Cov}(r_i, r_j)$     Correlation:  $\rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$

Var/Cov identities:

- $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$
- $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)$

Short-sell gain (sell at  $P_0$ , buy back at  $P_1$ ): profit =  $P_0 - P_1$ . Rate of return =  $(P_0 - P_1)/P_0 = 1 - P_1/P_0 = -r_{\text{long}}$ . **Loss is unlimited.**

Portfolio of 2 Assets

Weights  $w_1 = \alpha$ ,  $w_2 = 1 - \alpha$  ( $\alpha \in \mathbb{R}$  if short-selling allowed;  $\alpha \in [0, 1]$  otherwise).

$$\mu_p = \alpha \mu_1 + (1 - \alpha) \mu_2$$

$$\sigma_p^2 = \alpha^2 \sigma_1^2 + (1 - \alpha)^2 \sigma_2^2 + 2\alpha(1 - \alpha) \rho_{12} \sigma_1 \sigma_2$$

Global Minimum-Variance Portfolio (GMVP)

$$\alpha^* = \frac{\sigma_2^2 - \rho_{12} \sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12} \sigma_1 \sigma_2}$$

$$(\sigma_p^2)^* = \frac{\sigma_1^2 \sigma_2^2 (1 - \rho_{12}^2)}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12} \sigma_1 \sigma_2}$$

If  $\rho = 1$ : feasible set is a straight line. If  $\rho = -1$ : V-shape (zero variance achievable). If  $-1 < \rho < 1$ : curved bullet shape.

Efficient frontier: upper half of feasible set (including GMVP).

Check: if  $\alpha^* \notin [0, 1]$  and short-selling not allowed, the GMVP is the asset with lower  $\sigma$ .

## 6. Forwards & Futures

### Forward Contract

Obligation to buy/sell at forward price  $F(t, T)$  at maturity  $T$ .

- Long payoff:  $S_T - F(t, T)$
- Short payoff:  $F(t, T) - S_T$
- Costs nothing to enter (no premium)

### Forward Pricing (No-Arbitrage)

#### Non-dividend-paying asset:

$$F(0, T) = S_0 e^{rT} \text{ (cts)} \quad F(0, T) = S_0(1 + r)^T \text{ (disc)}$$

**General  $t$ :**  $F(t, T) = S_t e^{r(T-t)}$

**With storage cost  $c$**  (lump sum at  $t = 0$ ):

$$F(0, T) = (S_0 + c) e^{rT}$$

**With continuous dividend yield  $q$ :**

$$F(0, T) = S_0 e^{(r-q)T}$$

Forward price does **not** depend on expected return  $\mu$  of the underlying — only on  $S_0$ ,  $r$ ,  $T$  (and carry costs). This is a no-arbitrage result, not an expectation.

### Arbitrage Construction

If  $F(0, T) > S_0 e^{rT}$ : short forward, buy asset (borrow  $S_0$ ).

Profit at  $T$ :  $F - S_0 e^{rT} > 0$ .

If  $F(0, T) < S_0 e^{rT}$ : long forward, short-sell asset (invest proceeds). Profit at  $T$ :  $S_0 e^{rT} - F > 0$ .

### Futures vs. Forwards

|              | Forward     | Futures   |
|--------------|-------------|-----------|
| Settlement   | At maturity | Daily MTM |
| Counterparty | Bilateral   | Exchange  |

Daily P&L long futures:  $\approx F_{t+1} - F_t$  (margin account).

### Margin Account

Initial margin deposited. If balance falls below **maintenance margin**, margin call issued (top up to initial margin).

## 7. Hedging

### Perfect Hedge (Forwards/Futures)

Enter opposite position to fix price and eliminate risk. Conditions for perfect futures hedge: (1) delivery date matches; (2) same underlying; (3) integer multiple of contract size;

(4) sufficient liquidity.

### Hedging Direction ( $\rho > 0$ )

| Obligation   | Futures action  |                                    |
|--------------|-----------------|------------------------------------|
| Buy asset A  | Short futures B | If $\rho < 0$ : reverse direction. |
| Sell asset A | Long futures B  |                                    |

### Minimum-Variance Hedge

Hedge  $W$  units of asset A using futures on asset B (correlated). Short  $h$  futures contracts.

Cash flow at  $T$ :  $y = -WS_T + (F_T - F_0)h$

$$\text{Var}(y) = W^2 \text{Var}(S_T) - 2W \text{Cov}(S_T, F_T)h + \text{Var}(F_T)h^2$$

#### Optimal hedge ratio:

$$h^* = W \cdot \frac{\text{Cov}(S_T, F_T)}{\text{Var}(F_T)} = W \cdot \frac{\rho_{S,F} \sigma_S}{\sigma_F}$$

#### Minimum variance:

$$\text{Var}^*(y) = W^2 \text{Var}(S_T)(1 - \rho_{S,F}^2)$$

Hedge effectiveness:  $\rho_{S,F}^2$ . Factor  $\sqrt{1 - \rho^2}$  measures residual risk.

$h^*$  is in *units*, not contracts. Number of contracts =  $h^*/(\text{contract size})$ . Round to integer for real problems.

## 8. Options

### Key Definitions

**Call:** right to *buy* at strike  $K$ . **Put:** right to *sell* at strike  $K$ .

Long = holder (pays premium); Short = writer (receives premium).

European: exercise only at  $T$ . American: exercise any time  $\leq T$ . (Assume European unless stated.)

### Payoffs & Profits

| Position   | Payoff at $T$       | Profit            |
|------------|---------------------|-------------------|
| Long call  | $\max(S_T - K, 0)$  | payoff $-Ce^{rT}$ |
| Short call | $-\max(S_T - K, 0)$ | payoff $+Ce^{rT}$ |
| Long put   | $\max(K - S_T, 0)$  | payoff $-Pe^{rT}$ |
| Short put  | $-\max(K - S_T, 0)$ | payoff $+Pe^{rT}$ |

$C, P$  = option premium (time-0 price). Time-value of premium at  $T$ :  $Ce^{rT}$  or  $Pe^{rT}$ .

### Bounds

**Upper bounds:**  $C \leq S_0 \quad P \leq Ke^{-rT}$

**Lower bounds:**

$$C \geq \max(0, S_0 - Ke^{-rT}) \quad P \geq \max(0, Ke^{-rT} - S_0)$$

**Put–Call Parity** (European, no dividends):

$$C + Ke^{-rT} = P + S_0$$

Proof: Portfolio A (long call + zero-coupon bond paying  $K$ ) and Portfolio C (long put + stock) have identical payoffs  $\Rightarrow$  same price by no-arbitrage.

### Trading Strategies

**Bull spread:** Long call  $K_1$  + Short call  $K_2$ ,  $K_1 < K_2$ . Profit if  $S_T \uparrow$ .

**Bear spread:** Short call  $K_1$  + Long call  $K_2$  *or* Long put  $K_2$  + Short put  $K_1$ . Profit if  $S_T \downarrow$ .

**Butterfly** ( $K_1 < K_2 < K_3$ ,  $K_1 + K_3 = 2K_2$ ): Long  $C_{K_1}$ , Short  $2C_{K_2}$ , Long  $C_{K_3}$ :

| $S_T$ range          | Payoff      |
|----------------------|-------------|
| $S_T \leq K_1$       | 0           |
| $K_1 < S_T \leq K_2$ | $S_T - K_1$ |
| $K_2 < S_T \leq K_3$ | $K_3 - S_T$ |
| $S_T > K_3$          | 0           |

volatility bet).

**Straddle:** Long call + Long put (same  $K, T$ ). Profit if  $|S_T - K|$  large.

**Covered call:** Long stock + Short call. **Protective put:** Long stock + Long put.

### Binomial Option Pricing

Single period:  $S \rightarrow uS$  or  $dS$ , risk-free rate  $r$ ,  $R = 1 + r$ . No-arb requires  $u > R > d$ .

**Risk-neutral probability:**  $q = \frac{R - d}{u - d}$ ,  $1 - q = \frac{u - R}{u - d}$

$$C = \frac{1}{R}[qC_u + (1 - q)C_d]$$

$C_u = \max(uS - K, 0)$ ,  $C_d = \max(dS - K, 0)$  for call;  $\max(K - \cdot, 0)$  for put.

**Multi-period:** backward induction. Each node same formula. **CRR:**  $u = e^{\sigma\sqrt{\Delta t}}$ ,  $d = 1/u$ .

### Replication Portfolio (Binomial)

Construct portfolio:  $x$  shares +  $b$  in risk-free cash. Payoffs:

$$\text{up: } ux + Rb = C_u \quad \text{down: } dx + Rb = C_d$$

Solve:

$$x = \frac{C_u - C_d}{u - d} \quad b = \frac{uC_d - dC_u}{R(u - d)}$$

$$C = x + b = \frac{qC_u + (1 - q)C_d}{R} \quad \checkmark$$

Arbitrage: if  $C > x + b$ : short call, long portfolio. If  $C < x + b$ : long call, short portfolio.

### Put–Call Parity (General Form)

Let  $d$  = discount factor to maturity. Slides notation:

$$C + dK = P + S_0$$

Discrete:  $d = (1 + r)^{-T}$ . Continuous:  $d = e^{-rT}$ . Rearrange to find arbitrage: if  $C + dK \neq P + S_0$ , one side is cheap.

### Black–Scholes (Intro Only)

Stock GBM:  $dS = \mu S dt + \sigma S dz$ . CRR  $\rightarrow$  B–S as  $\Delta t \rightarrow 0$ :  $u = e^{\sigma\sqrt{\Delta t}}$ ,  $d = 1/u$ .

### Exam Patterns (Midterm)

**Q1 type — PV algebra:** Set up PV equation, let  $v = (1 + r)^{-1}$  or  $x = (1 + r)^{-2k}$ , solve quadratic/polynomial. Answer in terms of  $k$ .

**Q2 type — Loan refinancing:** (i) Use annuity formula to find  $L/X$  (match  $L$  =PV of annuity). (ii) Find balance after  $m$  payments. Convert interest rate to *effective per period*. New annuity equation for  $Y$ . **Key trap:** Lump sum paid immediately, then first instalment one month later (ordinary annuity shifted by 1 month in timeline).

**Q3 type — Variable-coupon bond** +  $R \neq F$ : Split into 4 annuity blocks (10 semi-annual payments each at different rates). Write PV equation with  $P$  on both sides (since  $R = P + 20$ ), solve linear equation.

**Q4 type — Newton–Raphson for bond yield:** Set  $f(\lambda)$  = PV of cash flows at  $\lambda - P_{\text{market}}$ . Compute  $f(0)$  as a starting check, choose  $\lambda_0$  near expected answer, iterate 1–2 times for 4 d.p.

### Common Mistakes

- Using nominal rate directly in annuity formula without converting
- Forgetting that annuity-due = ordinary annuity  $\times (1 + r)$
- Coupon =  $cF/m$  (not  $cF$ ) for semi-annual bonds
- Duration: weight by PV, not raw cash flow
- Forward price: uses *risk-free* rate  $r$ , not expected return  $\mu$
- Put–call parity: discount factor is  $e^{-rT}$  (continuous) or  $(1 + r)^{-T}$  (discrete)